

# Fixed- $\beta$ $c$ -Recursion for the Sphere Ramond Channel

The large- $c$  block of the contracted Ramond algebra

Private working note

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## Abstract

We study one problem only: the sphere four-point chiral block  $\langle V_4^{\text{NS}}(\infty)R_3(1)R_2(z)V_1^{\text{NS}}(0) \rangle$ , factorized through a generic Ramond module. In the central-charge recursion all Ramond momenta  $\beta$  are held fixed, so  $h_R(c, \beta) = c/24 - \beta^2$  and  $G_0^2 = -\beta^2$ . The Ramond Kac determinant gives the moving  $c$ -plane poles, while the singular-vector norm and factorization of the two endpoint Ward forms give their residues directly. The regular block at  $c = \infty$  is a separate problem and is determined solely from the contracted algebra.

The nonzero super-Virasoro modes contract to free bosonic and fermionic oscillators, while  $D = L_0 - c/24$  and  $G_0$  remain finite. The N-R-R vertices are coherent in this basis, so the raw block must first be stripped of its universal  $O(c)$  factor. The finite remainder is the full four-point block of this current algebra: both contracted vertices are sewn over all  $a_n$ - and  $\psi_n$ -Fock excitations and the Ramond ground fibre.

The all-level sewing can be evaluated without guessing the displaced vertices. Decompose the Ramond module into orthogonal global  $\mathfrak{sl}_2$  families before taking the limit. The strict R-R intertwiner is the identity on the nonzero oscillator Fock space, and the Möbius transport of the pair of pants preserves the global family generated by the Ramond ground state. Every positive-level Ramond quasi-primary coupling is consequently  $O(c^{-1/2})$  after unit-norm normalization, so its sewn coefficient is  $O(c^{-1})$ . Only the ground global family survives. Its exact hypergeometric block and Euler's transformation give

$$\lim_{c \rightarrow \infty} (1-z)^{c/12} \mathcal{B}_c(z) = (1-z)^{h_1+h_4+\beta_2^2+\beta_3^2}.$$

After the explicitly defined elliptic change of normalization, it gives the all-order regular term

$$S_\beta(q) = \left( \frac{16q}{z(q)} \right)^{\beta^2} (1-z(q))^{h_1+h_4+1/16} \theta_3(q)^{4(h_1+h_4-\beta_2^2-\beta_3^2)+1/4}.$$

Direct PBW/Gram/Ward sewing through level five, performed along the fixed- $\beta$  family, reproduces this limit for all four chiral sign choices.

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## 1 The question

Consider the even chiral block

$$\left\langle V_4^{\text{NS}}(\infty) R_{\beta_3}(1) R_{\beta_2}(z) V_1^{\text{NS}}(0) \right\rangle \quad (1)$$

in the channel in which  $R_{\beta_2}(z) V_1^{\text{NS}}(0)$  produces a generic long Ramond representation of momentum  $\beta$ . The analytic data held fixed as  $c$  varies are

$$h_1, \quad h_4, \quad \beta_2, \quad \beta_3, \quad \beta, \quad \eta_2, \eta_3 \in \{+1, -1\}. \quad (2)$$

The Ramond weights therefore move with  $c$ :

$$h_{\text{R}}(c, \gamma) = \frac{c}{24} - \gamma^2, \quad \gamma \in \{\beta, \beta_2, \beta_3\}. \quad (3)$$

This is the central point of the construction. Since

$$G_0^2 = L_0 - \frac{c}{24}, \quad (4)$$

the internal ground space obeys  $G_0^2 = -\beta^2$ , independently of  $c$ . There is no moving level-zero pole. We work at  $\beta \neq 0$ ; the shortened  $\beta = 0$  quotient is a separate limit.

The construction has two logically independent ingredients. At finite  $c$ , the Ramond Kac determinant and singular-vector factorization determine the complete pole part directly on the fixed- $\beta$   $c$ -plane. At  $c = \infty$ , the contracted Ramond algebra determines the regular part. The nontrivial large- $c$  problem is

$$\boxed{\text{compute the regular } c \rightarrow \infty \text{ block at fixed } \beta, \beta_2, \beta_3, h_1, h_4.} \quad (5)$$

## 2 Definition of the block

Let  $\nu_i$  denote the NS highest-weight state of weight  $h_i$ :

$$L_0 \nu_i = h_i \nu_i, \quad L_{n>0} \nu_i = G_{r>0} \nu_i = 0, \quad r \in \mathbb{Z} + \frac{1}{2}. \quad (6)$$

Let  $\mathcal{W}_{c,\gamma}$  be a Ramond Verma module. Its even ground state  $w_\gamma^+$  is fixed by

$$L_0 w_\gamma^+ = h_R(c, \gamma) w_\gamma^+, \quad L_{n>0} w_\gamma^+ = G_{n>0} w_\gamma^+ = 0, \quad (-1)^F w_\gamma^+ = w_\gamma^+, \quad \langle w_\gamma^+ | w_\gamma^+ \rangle = 1. \quad (7)$$

The other ground state is generated by  $G_0 w_\gamma^+$ . We do not normalize it separately; its norm is proportional to  $h_R - c/24 = -\gamma^2$ .

An even internal PBW basis state at level  $n$  is denoted

$$w_{\beta,I}^+ = L_{-m_1} \cdots L_{-m_p} G_{-k_1} \cdots G_{-k_q} G_0^\epsilon w_\beta^+, \quad (8)$$

where  $m_a, k_b \in \mathbb{Z}_{>0}$ , the fermionic mode numbers  $k_b$  are distinct, and

$$|I| = \sum_a m_a + \sum_b k_b = n, \quad q + \epsilon = 0 \pmod{2}, \quad \epsilon \in \{0, 1\}. \quad (9)$$

Thus the superscript  $+$  on  $w_{\beta,I}^+$  denotes the total even fermion-parity sector, while  $I$  is shorthand for the complete PBW label  $(m_1, \dots, m_p; k_1, \dots, k_q; \epsilon)$ .

In Eq. (13) below,  $\nu_4 \in \mathcal{V}_{h_4}^{\text{NS}}$ ,  $w_{\beta_3}^+ \in \mathcal{W}_{c,\beta_3}$ , and  $w_{\beta,I}^+ \in \mathcal{W}_{c,\beta}$ . Likewise,  $\nu_1 \in \mathcal{V}_{h_1}^{\text{NS}}$  and  $w_{\beta_2}^+ \in \mathcal{W}_{c,\beta_2}$ . At integer level  $n$ , let  $B_{R,n}(c, \beta)$  denote the Gram matrix of the basis (8).

Let the unnormalized chiral three-point forms have domains

$$\begin{aligned} \varrho_{NR}^{(\eta_3)} : \mathcal{V}_{h_4}^{\text{NS}} \times \mathcal{W}_{c,\beta_3} \times \mathcal{W}_{c,\beta} &\longrightarrow \mathbb{C}, \\ \varrho_{RN}^{(\eta_2)} : \mathcal{W}_{c,\beta} \times \mathcal{W}_{c,\beta_2} \times \mathcal{V}_{h_1}^{\text{NS}} &\longrightarrow \mathbb{C}. \end{aligned} \quad (10)$$

Here  $\eta_2, \eta_3 = \pm 1$  choose one of the two even Ramond chiral structures. Their ground-state matrix elements are the chiral three-point coefficients

$$C_L^{(\eta_3)} := \varrho_{NR}^{(\eta_3)}(\nu_4, w_{\beta_3}^+, w_\beta^+), \quad C_R^{(\eta_2)} := \varrho_{RN}^{(\eta_2)}(w_\beta^+, w_{\beta_2}^+, \nu_1). \quad (11)$$

They are not set to one. The Ward identities determine descendant matrix elements only relative to these coefficients. Away from their zeros, the normalized three-forms used to define the conformal block are

$$\rho_{NR}^{(\eta_3)} := \frac{\varrho_{NR}^{(\eta_3)}}{C_L^{(\eta_3)}}, \quad \rho_{RN}^{(\eta_2)} := \frac{\varrho_{RN}^{(\eta_2)}}{C_R^{(\eta_2)}}.$$

Thus the normalized ground values are one by definition, while Eq. (11) retains the actual three-point data. At a zero of either coefficient, the normalized block is understood by analytic continuation after this formal factorization.

The local series is

$$\mathcal{F}_\beta^{\eta_3, \eta_2}(z) = z^{h_R(\beta) - h_R(\beta_2) - h_1} \mathcal{B}_\beta^{\eta_3, \eta_2}(c; z), \quad \mathcal{B}_\beta(c; z) = \sum_{n=0}^{\infty} B_n(c) z^n, \quad (12)$$

where the  $(\eta_3, \eta_2)$  labels on  $\mathcal{B}_\beta$  and  $B_n$  are suppressed on the second equality, and

$$B_n(c) = \sum_{|I|=|J|=n} \rho_{NR}^{(\eta_3)}(\nu_4, w_{\beta_3}^+, w_{\beta,I}^+) (B_{R,n}^{-1})^{IJ} \rho_{RN}^{(\eta_2)}(w_{\beta,J}^+, w_{\beta_2}^+, \nu_1). \quad (13)$$

The convention in Eq. (13) is the following.

1. The radial ordering is

$$V_1^{\text{NS}}(0), \quad R_{\beta_2}(z), \quad R_{\beta_3}(1), \quad V_4^{\text{NS}}(\infty).$$

After the overall power of  $z$  in Eq. (12) is extracted, each normalized three-form in Eq. (13) is evaluated at unit insertion position.

2. In  $\rho_{NR}(\nu_4, w_{\beta_3}^+, w_{\beta, I}^+)$ , the first argument is the outgoing NS bra, the second is the inserted external Ramond ground state, and the third is the incoming internal Ramond state. In  $\rho_{RN}(w_{\beta, J}^+, w_{\beta_2}^+, \nu_1)$ , the first argument is the outgoing internal Ramond state, the second is the inserted external Ramond ground state, and the third is the incoming NS ket. Hence these are the ordered  $NR$  and  $RN$  forms, not an  $RR$  form with an NS insertion. Equivalently, in matrix-element notation,

$$\begin{aligned} \rho_{NR}^{(\eta_3)}(\nu_4, w_{\beta_3}^+, w_{\beta, I}^+) &= \frac{1}{C_L^{(\eta_3)}} \langle \nu_4 | R_{\beta_3}^{(\eta_3)}(1) | w_{\beta, I}^+ \rangle, \\ \rho_{RN}^{(\eta_2)}(w_{\beta, J}^+, w_{\beta_2}^+, \nu_1) &= \frac{1}{C_R^{(\eta_2)}} \langle w_{\beta, J}^+ | R_{\beta_2}^{(\eta_2)}(1) | \nu_1 \rangle. \end{aligned} \quad (14)$$

Here the numerators are the unnormalized chiral three-point matrix elements, and  $R_{\beta_i}^{(\eta_i)}$  denotes the Ramond intertwiner selected by the chiral-structure label  $\eta_i$ . The superscript does not introduce an additional external state.

3. The Gram matrix and its inverse are

$$B_{R,n}[I, J] = \langle w_{\beta, I}^+ | w_{\beta, J}^+ \rangle, \quad \sum_J B_{R,n}[I, J] (B_{R,n}^{-1})^{JK} = \delta_I^K, \quad (15)$$

with radial BPZ pairing  $L_n^\dagger = L_{-n}$  and  $G_n^\dagger = G_{-n}$ . The indices  $I, J$  run over the complete even PBW basis (8) at the same integer level  $n$ , including the unnormalized  $G_0 w_\beta^+$  ground direction when required by parity. Thus  $(B_{R,n}^{-1})^{IJ}$  is precisely the internal Ramond sewing propagator.

4. The labels  $\eta_3, \eta_2 = \pm 1$  choose the two even chiral Ward solutions at the left and right vertices. They are not fermion-parity labels on  $w^+$ , and they are not the nonchiral  $R^\pm$  field labels. The corresponding unnormalized ground matrix elements are  $C_L^{(\eta_3)}$  and  $C_R^{(\eta_2)}$  in Eq. (11).
5. Equation (13) uses the divided forms  $\rho = \varrho/C$ , so it defines the normalized holomorphic conformal block, with  $B_0(c) = 1$ . The unnormalized holomorphic contribution in this channel is  $C_L^{(\eta_3)} C_R^{(\eta_2)} \mathcal{F}_\beta^{\eta_3, \eta_2}(z)$ . Hence the three-point coefficients have been factored out, not set to one. Constructing a full correlator additionally requires the antiholomorphic blocks and their allowed left–right pairing. Any  $c$ -dependence, zero, or pole of  $C_L C_R$  is dynamical CFT data and is not part of the universal representation-theoretic  $c$ -recursion.

These conventions completely specify the block used below.

### 3 The $c$ -plane poles from the Ramond Kac determinant

Use the ordinary central charge

$$c(b) = \frac{3}{2} + 3(b + b^{-1})^2. \quad (16)$$

For  $r, s \geq 1$ ,  $r + s$  odd, the Ramond degenerate momentum is

$$\beta_{r,s}(b) = \frac{rb + s/b}{2\sqrt{2}}. \quad (17)$$

At fixed internal  $\beta$ , a  $c$ -pole is obtained by solving

$$\beta = \beta_{r,s}(b_*), \quad c_* = c(b_*). \quad (18)$$

Choose  $r > s$  and keep the two roots

$$b_{r,s}^{(\alpha)}(\beta) = \frac{\sqrt{2}\beta + \alpha\sqrt{2\beta^2 - rs}}{r}, \quad c_{r,s}^{(\alpha)}(\beta) = c(b_{r,s}^{(\alpha)}(\beta)), \quad \alpha = \pm. \quad (19)$$

This counts each generic pole once: the descriptions  $(s, r, 1/b)$  and  $-\beta = \beta_{r,s}(-b)$  give the same  $c$ -values and must not be added again.

For later use define the Jacobian and shifted Ramond momentum

$$\begin{aligned} J_{r,s}^{(\alpha)}(\beta) &:= \left. \frac{\partial \beta_{r,s}}{\partial c} \right|_{b=b_{r,s}^{(\alpha)}} = \left. \frac{r - s/b^2}{12\sqrt{2}(b + b^{-1})(1 - b^{-2})} \right|_{b=b_{r,s}^{(\alpha)}}, \\ \beta'_{r,s}(\beta) &:= \frac{(-1)^s}{2\sqrt{2}} \left( rb - \frac{s}{b} \right) \Big|_{b=b_{r,s}^{(\alpha)}}. \end{aligned} \quad (20)$$

The second line is characterized by  $h_R(c_*, \beta'_{r,s}) = h_R(c_*, \beta_{r,s}) + rs/2$ .

We now make the two representation-theoretic factors explicit. With the singular vector normalized by  $\chi_{r,s}^+ = L_{-1}^{rs/2} w_{r,s}^+ + \dots$ , the inverse Ramond null norm is

$$A_{r,s}^R(b) = -2^{rs-\frac{3}{2}} \prod_{\substack{m=1-r, \dots, r \\ n=1-s, \dots, s \\ m+n \equiv 0 \pmod{2} \\ (m,n) \neq (0,0), (r,s)}} \left( mb + \frac{n}{b} \right)^{-1}. \quad (21)$$

The even-sublattice condition in Eq. (21) is fixed by the defining inverse-norm limit and is already verified by the exact level-one  $(r, s) = (2, 1)$  Gram matrix.

For an NS weight  $h$ , introduce a momentum branch

$$Q = b + b^{-1}, \quad h = \frac{Q^2 - \lambda(h; b)^2}{8}. \quad (22)$$

The branch must be chosen once and continued consistently in  $b$ . The implementation uses  $\lambda(h; b) = -\sqrt{Q^2 - 8h}$ . Define

$$p_k = 1 - r + 2k, \quad q_\ell = 1 - s + 2\ell, \quad \mu_{k\ell}(b) = \frac{p_k b + q_\ell / b}{2\sqrt{2}}, \quad \begin{matrix} 0 \leq k \leq r-1 \\ 0 \leq \ell \leq s-1 \end{matrix}. \quad (23)$$

Then the N-R-R fusion polynomial appearing at either end of the internal line is

$$\begin{aligned} P_{r,s}^\eta(\beta_R, h; b) &= \prod_{\substack{0 \leq k < r, \ 0 \leq \ell < s \\ k+\ell \text{ even}}} \left[ \frac{\lambda(h; b)}{2\sqrt{2}} - \eta\beta_R - \mu_{k\ell}(b) \right] \\ &\times \prod_{\substack{0 \leq k < r, \ 0 \leq \ell < s \\ k+\ell \text{ odd}}} \left[ \frac{\lambda(h; b)}{2\sqrt{2}} + \eta\beta_R - \mu_{k\ell}(b) \right], \quad \eta = \pm 1. \end{aligned} \quad (24)$$

These products are precisely the singular-vector factorization coefficients of the normalized  $NR$  and  $RN$  three-forms; the three-point coefficients  $C_L C_R$  remain outside the conformal block.

Using Eqs. (21) and (24), define the complete  $c$ -plane residue multiplier

$$\mathcal{K}_{r,s}^{(\alpha;\eta_3,\eta_2)}(\beta) := -\frac{16^{rs/2} A_{r,s}^R(b_\alpha)}{J_{r,s}^{(\alpha)}(\beta)} P_{r,s}^{\eta_3}(\beta_3, h_4; b_\alpha) P_{r,s}^{\eta_2}(\beta_2, h_1; b_\alpha), \quad b_\alpha = b_{r,s}^{(\alpha)}(\beta). \quad (25)$$

Thus  $\mathcal{K}$  is an explicit finite product; no additional residue coefficient is left unspecified. The residue of the normalized elliptic conformal block is then

$$\boxed{\text{Res}_{c=c_{r,s}^{(\alpha)}(\beta)} \mathcal{H}_\beta^{\eta_3,\eta_2}(c; q) = q^{rs/2} \mathcal{K}_{r,s}^{(\alpha;\eta_3,\eta_2)}(\beta) \mathcal{H}_{\beta'_{r,s}^{(\alpha)}(\beta)}^{\eta_3,\eta_2}(c_{r,s}^{(\alpha)}(\beta); q).} \quad (26)$$

Equivalently, for the local reduced block  $\mathcal{B}_\beta(c; z)$  of Eq. (12),

$$\boxed{\text{Res}_{c=c_{r,s}^{(\alpha)}(\beta)} \mathcal{B}_\beta^{\eta_3,\eta_2}(c; z) = \left(\frac{z}{16}\right)^{rs/2} \mathcal{K}_{r,s}^{(\alpha;\eta_3,\eta_2)}(\beta) \mathcal{B}_{\beta'_{r,s}^{(\alpha)}(\beta)}^{\eta_3,\eta_2}(c_{r,s}^{(\alpha)}(\beta); z).} \quad (27)$$

The factor  $16^{-rs/2}$  removes the  $16^{rs/2}$  already included in  $\mathcal{K}$ . Thus the local descendant series starts its singular-vector tail at  $z^{rs/2}$ , as it must.

Indeed,

$$\beta - \beta_{r,s}(c) = -J_{r,s}^{(\alpha)}(\beta)(c - c_{r,s}^{(\alpha)}(\beta)) + O((c - c_{r,s}^{(\alpha)}(\beta))^2),$$

which explains both the minus sign and the inverse Jacobian in Eq. (25). The endpoint labels  $(\eta_3, \eta_2)$  are unchanged on this canonical positive- $\beta$  sheet. Equations (25)–(27) assume a generic simple pole:  $J_{r,s}^{(\alpha)} \neq 0$  and no collision with another Kac pole. Ramified or colliding loci require a separate limiting prescription.

No new singular-vector calculation is required. The same residue formula holds for the oscillator-normalized block  $\hat{\mathcal{H}}$  defined in Eq. (85): the factor  $\mathcal{E}_R(c, q)$  is holomorphic and nonzero in  $c$ , and its value at the pole cancels against the identical factor multiplying the shifted block.

## 4 The contracted Ramond algebra

### 4.1 Finite zero modes

Define

$$D = L_0 - \frac{c}{24}. \quad (28)$$

On the ground doublet,

$$D|\beta, a\rangle = -\beta^2|\beta, a\rangle, \quad G_0^2 = D. \quad (29)$$

Both  $D$  and  $G_0$  remain unscaled.

### 4.2 Normalized nonzero modes

For  $n \geq 1$ , introduce

$$a_n = \sqrt{\frac{12}{cn^3}} L_n, \quad a_n^\dagger = \sqrt{\frac{12}{cn^3}} L_{-n}, \quad \psi_n = \sqrt{\frac{3}{cn^2}} G_n, \quad \psi_n^\dagger = \sqrt{\frac{3}{cn^2}} G_{-n}. \quad (30)$$

On fixed-level states the  $c \rightarrow \infty$  brackets are

$$\begin{aligned} [a_n, a_m^\dagger] &= \delta_{nm}, & \{\psi_n, \psi_m^\dagger\} &= \delta_{nm}, \\ [a_n, a_m] &= [a_n^\dagger, a_m^\dagger] = 0, & \{\psi_n, \psi_m\} &= \{\psi_n^\dagger, \psi_m^\dagger\} = 0. \end{aligned} \quad (31)$$

The surviving zero-mode action is

$$\begin{aligned} [D, a_n] &= -na_n, & [D, a_n^\dagger] &= na_n^\dagger, \\ [D, \psi_n] &= -n\psi_n, & [D, \psi_n^\dagger] &= n\psi_n^\dagger, \\ [G_0, a_n] &= -\sqrt{n}\psi_n, & [G_0, a_n^\dagger] &= \sqrt{n}\psi_n^\dagger, \\ \{G_0, \psi_n\} &= \sqrt{n}a_n, & \{G_0, \psi_n^\dagger\} &= \sqrt{n}a_n^\dagger. \end{aligned} \quad (32)$$

Together with  $G_0^2 = D$ , this defines

$$\mathfrak{A}_{R,\infty} = (\mathfrak{h}_{\text{bos}} \oplus \mathfrak{h}_{\text{fer}}) \rtimes \langle D, G_0 \rangle. \quad (33)$$

It is an infinite free super-oscillator algebra with a common Ramond supercharge.

### 4.3 Action on states and field intertwiners

The commutators above can be integrated to an explicit action on the contracted Ramond Fock module. Let  $\gamma_\beta$  be the restriction of  $G_0$  to the ground doublet, so that  $\gamma_\beta^2 = -\beta^2$ . On the graded tensor product of this doublet with the oscillator Fock space,

$$\begin{aligned} D &= -\beta^2 + \sum_{n \geq 1} n (a_n^\dagger a_n + \psi_n^\dagger \psi_n), \\ G_0 &= \gamma_\beta + \sum_{n \geq 1} \sqrt{n} (a_n^\dagger \psi_n + \psi_n^\dagger a_n). \end{aligned} \quad (34)$$

The graded tensor product is essential:  $\gamma_\beta$  anticommutes with the fermionic oscillators. Equation (34) then gives  $G_0^2 = D$  and the brackets Eq. (32). For example, if  $|0, g\rangle$  is an oscillator vacuum with an open ground index,

$$\begin{aligned} G_0 a_n^\dagger |0, g\rangle &= a_n^\dagger \gamma_\beta |0, g\rangle + \sqrt{n} \psi_n^\dagger |0, g\rangle, \\ G_0 \psi_n^\dagger |0, g\rangle &= -\psi_n^\dagger \gamma_\beta |0, g\rangle + \sqrt{n} a_n^\dagger |0, g\rangle. \end{aligned} \quad (35)$$

Thus  $G_0$  does not act as a scalar on an excited state: it exchanges the bosonic and fermionic excitation of each mode number.

It is equally useful to record the action on fields. First use the R–R radial presentation, in which an even NS primary  $\mathcal{V}_h(z)$  and its odd upper component  $\mathcal{U}_h(z) = (G_{-1/2} \mathcal{V}_h)(z)$  are intertwiners between two Ramond modules. Superconformal covariance gives, for  $m \in \mathbb{Z}$ ,

$$\begin{aligned} L_m^{\text{out}} \mathcal{V}_h - \mathcal{V}_h L_m^{\text{in}} &= z^m (z \partial_z + (m+1)h) \mathcal{V}_h, \\ G_m^{\text{out}} \mathcal{V}_h - \mathcal{V}_h G_m^{\text{in}} &= z^{m+1/2} \mathcal{U}_h, \\ G_m^{\text{out}} \mathcal{U}_h + \mathcal{U}_h G_m^{\text{in}} &= z^{m-1/2} (z \partial_z + (2m+1)h) \mathcal{V}_h. \end{aligned} \quad (36)$$

In particular, the unscaled generators act by

$$D^{\text{out}} \mathcal{V}_h - \mathcal{V}_h D^{\text{in}} = (z \partial_z + h) \mathcal{V}_h,$$

$$\begin{aligned}
G_0^{\text{out}} \mathcal{V}_h - \mathcal{V}_h G_0^{\text{in}} &= z^{1/2} \mathcal{U}_h, \\
G_0^{\text{out}} \mathcal{U}_h + \mathcal{U}_h G_0^{\text{in}} &= z^{-1/2} (z \partial_z + h) \mathcal{V}_h.
\end{aligned} \tag{37}$$

For  $n \geq 1$ , the normalized nonzero-mode relations are

$$\begin{aligned}
a_n^{\text{out}} \mathcal{V}_h - \mathcal{V}_h a_n^{\text{in}} &= \frac{2\sqrt{3}}{\sqrt{c} n^{3/2}} z^n (z \partial_z + (n+1)h) \mathcal{V}_h, \\
\psi_n^{\text{out}} \mathcal{V}_h - \mathcal{V}_h \psi_n^{\text{in}} &= \frac{\sqrt{3}}{\sqrt{c} n} z^{n+1/2} \mathcal{U}_h, \\
\psi_n^{\text{out}} \mathcal{U}_h + \mathcal{U}_h \psi_n^{\text{in}} &= \frac{\sqrt{3}}{\sqrt{c} n} z^{n-1/2} (z \partial_z + (2n+1)h) \mathcal{V}_h.
\end{aligned} \tag{38}$$

The corresponding creation-mode formulae follow by setting  $m = -n$  in Eq. (36). Since  $h$  is fixed, every right-hand side in Eq. (38) vanishes in the strict contraction. Hence the R–R presentation of the bottom NS field intertwines all nonzero  $a_n, \psi_n$  oscillators diagonally, while its  $D, G_0$  action remains nontrivial.

This statement must not be applied directly to the N–R–R and R–R–N radial vertices in the sphere block. There the inserted field is Ramond, the supercurrent has a square-root branch, and there is no ordinary single-mode commutator. The exact action is the generalized Ramond Ward system. To display it compactly, set

$$C_p(x) = \binom{x}{p}, \quad \varepsilon(\xi, \eta) = -i(-1)^{|\xi|+|\eta|+1},$$

and write  $\rho = \rho_{NR}(\xi, \eta_3, \eta; z)$ . For  $n \in \mathbb{Z}$ , the two contour expansions are

$$\begin{aligned}
&\sum_{p \geq 0} C_p \left( n + \frac{1}{2} \right) z^{n+1/2-p} \rho_{NR}(\xi, G_p \eta_3, \eta; z) \\
&= \sum_{p \geq 0} C_p \left( \frac{1}{2} \right) (-z)^p \rho_{NR}(G_{p-n-1/2} \xi, \eta_3, \eta; z) \\
&\quad + \varepsilon(\xi, \eta) \sum_{p \geq 0} C_p \left( \frac{1}{2} \right) (-1)^p z^{1/2-p} \rho_{NR}(\xi, \eta_3, G_{n+p} \eta; z),
\end{aligned} \tag{39}$$

$$\begin{aligned}
&\sum_{p \geq 0} C_p \left( \frac{1}{2} \right) z^{1/2-p} \rho_{NR}(\xi, G_{p-n} \eta_3, \eta; z) \\
&= \sum_{p \geq 0} C_p \left( \frac{1}{2} - n \right) (-z)^p \rho_{NR}(G_{p+n-1/2} \xi, \eta_3, \eta; z) \\
&\quad + \varepsilon(\xi, \eta) \sum_{p \geq 0} C_p \left( \frac{1}{2} - n \right) (-1)^{n+p} z^{1/2-n-p} \rho_{NR}(\xi, \eta_3, G_p \eta; z).
\end{aligned} \tag{40}$$

These generalized Ramond Ward identities follow directly by deforming the supercurrent contour around the three punctures; the R–N relation is their BPZ/Hermitian counterpart. On fixed-level states the apparently infinite sums truncate. If  $\eta_3 = w_{\beta_3}^+$  is a Ramond ground state and  $\xi = \nu_4$  is the external NS primary, the  $n = 0$  case of Eq. (40) isolates the field zero-mode action:

$$\begin{aligned}
z^{1/2} \rho_{NR}(\nu_4, G_0 w_{\beta_3}^+, \eta; z) &= \rho_{NR}(G_{-1/2} \nu_4, w_{\beta_3}^+, \eta; z) \\
&\quad + \varepsilon(\nu_4, \eta) \sum_{p \geq 0} C_p \left( \frac{1}{2} \right) (-1)^p z^{1/2-p} \rho_{NR}(\nu_4, w_{\beta_3}^+, G_p \eta; z).
\end{aligned} \tag{41}$$



Thus the  $G_0$  action on the Ramond field, the upper NS component, and the  $G_p$  action on the internal state form one closed system.

To obtain the contracted vertex amplitudes, substitute

$$G_n = n\sqrt{\frac{c}{3}}\psi_n, \quad G_{-n} = n\sqrt{\frac{c}{3}}\psi_n^\dagger, \quad n \geq 1, \quad (42)$$

into Eqs. (39) and (40), keep  $G_0$  unscaled, perform the bosonic coherent displacement, and only then take  $c \rightarrow \infty$ . This gives a triangular determination of the  $\Gamma_L, \Gamma_R$  appearing in Eq. (54).

There is a complementary geometric way to organize the same calculation. Reordering an N–R–R three-point function into the R–R presentation changes the local coordinate and the spin frame. For example, the Möbius map

$$f(x) = \frac{x}{x-1} \quad (43)$$

fixes 0 and exchanges 1 with  $\infty$ . Its transport operator  $\mathcal{U}_f$  is fixed by

$$\begin{aligned} \mathcal{U}_f^{-1}T(x)\mathcal{U}_f &= f'(x)^2T(f(x)) + \frac{c}{12}\{f, x\}, \\ \mathcal{U}_f^{-1}G(x)\mathcal{U}_f &= \varsigma_f f'(x)^{3/2}G(f(x)), \end{aligned} \quad (44)$$

where  $\varsigma_f = \pm 1$  is the chosen spin lift; the Schwarzian vanishes for Eq. (43). Expanding Eq. (44) at  $x = 0$  gives the explicit mode transport

$$\begin{aligned} \mathcal{U}_f^{-1}L_m\mathcal{U}_f &= \sum_{j=m}^{\infty} (-1)^j \binom{1-m}{j-m} L_j, \\ \mathcal{U}_f^{-1}G_m\mathcal{U}_f &= \varsigma_f \sum_{j=m}^{\infty} (-1)^j \binom{\frac{1}{2}-m}{j-m} G_j. \end{aligned} \quad (45)$$

The sums are formal puncture expansions and are applied coefficientwise. For a bosonic creation mode the first sum terminates, and Eq. (45) becomes

$$\begin{aligned} \mathcal{U}_f^{-1}a_N^\dagger\mathcal{U}_f &= \sum_{r=1}^N (-1)^r \binom{N+1}{N-r} \left(\frac{r}{N}\right)^{3/2} a_r^\dagger - \frac{1}{N^{3/2}}a_1 \\ &\quad + \sqrt{c} \frac{N+1}{4\sqrt{3}N^{3/2}} + \frac{2\sqrt{3}(N+1)}{\sqrt{c}N^{3/2}}D. \end{aligned} \quad (46)$$

The third term is precisely the coherent source in Eq. (56). The fermionic creation mode instead undergoes a Bogoliubov transformation:

$$\begin{aligned} \mathcal{U}_f^{-1}\psi_N^\dagger\mathcal{U}_f &= \varsigma_f \left[ \sum_{r=1}^N (-1)^r \binom{N+\frac{1}{2}}{N-r} \frac{r}{N} \psi_r^\dagger \right. \\ &\quad \left. + \sum_{s \geq 1} (-1)^s \binom{N+\frac{1}{2}}{N+s} \frac{s}{N} \psi_s + \frac{\sqrt{3}}{\sqrt{c}N} \binom{N+\frac{1}{2}}{N} G_0 \right]. \end{aligned} \quad (47)$$

Finally, the ground supercharge itself transforms as

$$\mathcal{U}_f^{-1}G_0\mathcal{U}_f = \varsigma_f \left[ G_0 + \sqrt{\frac{c}{3}} \sum_{s \geq 1} (-1)^s s \binom{\frac{1}{2}}{s} \psi_s \right]. \quad (48)$$

Thus the bosonic transport is affine, the fermionic transport mixes creation and annihilation modes, and the Ramond zero mode remains tied to the fermionic annihilation modes. These transformations, together with the finite ground-fibre intertwiner in Eq. (37), reduce the regular block to a coherent/Bogoliubov Fock-space sewing problem.

Although the NS field is transparent to the strict nonzero oscillators in Eq. (38),  $\mathcal{U}_f$  has a nontrivial coherent large- $c$  limit because  $L_0 = c/24 + D$ . Consequently the simple R–R intertwiner and the original coherent N–R–R vertex are compatible rather than contradictory. Evaluating this transported current-algebra intertwiner is an equivalent route to the regular block.

The mechanism is already visible at level one. Since  $f'(0) = -1$  and  $f''(0) = -2$ , the descendant transforms, relative to the primary transport, as

$$\mathcal{U}_f L_{-1} w_\beta = (-L_{-1} + 2h_R(\beta)) \mathcal{U}_f w_\beta. \quad (49)$$

In the R–R presentation the fixed-weight NS insertion gives

$$\frac{\rho_{RR}(w_{\beta_3}^+, \nu_4, L_{-1} w_\beta^+)}{\rho_{RR}(w_{\beta_3}^+, \nu_4, w_\beta^+)} = h_R(\beta) + h_4 - h_R(\beta_3).$$

After Eq. (49), the coefficient in the original N–R–R presentation is

$$-[h_R(\beta) + h_4 - h_R(\beta_3)] + 2h_R(\beta) = h_R(\beta) + h_R(\beta_3) - h_4, \quad (50)$$

which is exactly Eq. (55). Its  $O(c)$  part therefore comes from local-coordinate transport, not from a nontrivial strict commutator of  $a_n$  with the fixed-weight NS field.

#### 4.4 The regular term is the full current-algebra block

The contracted algebra in Eq. (33) is the intermediate-state algebra of the regular term. To say that the regular term is its four-point block means that one must sew the two contracted three-point vertices over the *complete even Fock sector*, including every bosonic oscillator  $a_n^\dagger$ , every fermionic oscillator  $\psi_n^\dagger$ , and the open Ramond ground fibre.

More explicitly, order the ground zero mode to the right and introduce the even Fock basis

$$|\mathbf{r}, \boldsymbol{\epsilon}; \sigma\rangle = \prod_{n \geq 1} (a_n^\dagger)^{r_n} \prod_{n \geq 1} (\psi_n^\dagger)^{\epsilon_n} G_0^\sigma w_\beta^+, \quad \sigma + \sum_{n \geq 1} \epsilon_n = 0 \pmod{2}, \quad (51)$$

where only finitely many entries are nonzero,  $r_n \in \mathbb{Z}_{\geq 0}$ , and  $\epsilon_n, \sigma \in \{0, 1\}$ . Its level is

$$N(\mathbf{r}, \boldsymbol{\epsilon}) = \sum_{n \geq 1} n(r_n + \epsilon_n). \quad (52)$$

With  $\langle w_\beta^+ | w_\beta^+ \rangle = 1$ , the strict contracted Gram matrix in this ordered basis is diagonal:

$$\langle \mathbf{r}, \boldsymbol{\epsilon}; \sigma | \mathbf{r}', \boldsymbol{\epsilon}'; \sigma' \rangle = \delta_{\mathbf{r}\mathbf{r}'} \delta_{\boldsymbol{\epsilon}\boldsymbol{\epsilon}'} \delta_{\sigma\sigma'} \left( \prod_{n \geq 1} r_n! \right) (-\beta^2)^\sigma. \quad (53)$$

Let  $\Gamma_L^{\eta_3}(\mathbf{r}, \boldsymbol{\epsilon}; \sigma)$  and  $\Gamma_R^{\eta_2}(\mathbf{r}, \boldsymbol{\epsilon}; \sigma)$  denote the finite left and right vertex amplitudes obtained from the normalized N–R–R and R–R–N Ward identities after the coherent displacement described below. The current-algebra block is then, by definition,

$$\mathcal{B}_{\text{cur}, \beta}^{\eta_3, \eta_2}(z) = \sum_{\substack{\mathbf{r}, \boldsymbol{\epsilon}, \sigma \\ \sigma + \sum_n \epsilon_n \equiv 0 \pmod{2}}} z^{N(\mathbf{r}, \boldsymbol{\epsilon})} \frac{\Gamma_L^{\eta_3}(\mathbf{r}, \boldsymbol{\epsilon}; \sigma) \Gamma_R^{\eta_2}(\mathbf{r}, \boldsymbol{\epsilon}; \sigma)}{\left( \prod_{n \geq 1} r_n! \right) (-\beta^2)^\sigma}. \quad (54)$$

Here  $\Gamma_L(\mathbf{0}, \mathbf{0}; 0) = \Gamma_R(\mathbf{0}, \mathbf{0}; 0) = 1$ , as follows from the normalized three-point forms. Equation (54) is written for generic  $\beta \neq 0$ ; the  $\beta = 0$  ground fibre requires a separate limiting prescription. Thus the summation over the  $a_n$  and  $\psi_n$  excitations is not an optional correction to the seed: it *is* the regular term. Equation (54) is not a vacuum character, because the two  $\Gamma$ 's retain the external weights, external Ramond momenta, signs  $\eta_3, \eta_2$ , and all vertex Ward data. The contracted algebra fixes the propagator Eq. (53); the contracted Ward identities fix the two vertices.

There is one important order-of-limits issue in defining the finite  $\Gamma$ 's. The N–R–R vertices become coherent. For example, the Virasoro Ward identities give

$$\begin{aligned}\frac{\rho_L(L_{-n}w_\beta)}{\rho_L(w_\beta)} &= h_R(\beta) + nh_R(\beta_3) - h_4, \\ \frac{\rho_R(L_{-n}w_\beta)}{\rho_R(w_\beta)} &= h_R(\beta) + nh_R(\beta_2) - h_1.\end{aligned}\tag{55}$$

Consequently,

$$\frac{\rho_{L,R}(a_n^\dagger w_\beta)}{\rho_{L,R}(w_\beta)} = \sqrt{c} \frac{n+1}{4\sqrt{3}n^{3/2}} + O(c^{-1/2}).\tag{56}$$

Thus the raw normalized vertex amplitudes do not possess a finite mode-by-mode limit before the coherent piece is removed. Subleading mode-changing brackets can combine with the  $O(\sqrt{c})$  sources and leave finite contributions to the displaced vertices. Consequently, the strict diagonal Gram matrix must not be inserted before the vertices have been displaced. The correct operation is:

1. rewrite the exact  $NR/RN$  Ward identities in the normalized variables;
2. perform the coherent displacement;
3. take the coefficientwise large- $c$  limit to obtain the finite amplitudes  $\Gamma_L$  and  $\Gamma_R$ ;
4. perform the complete Fock-space sewing in Eq. (54).

Once the two displaced vertices are known, no finite- $c$  descendant data remain: the regular term is computed entirely as the four-point block of  $\mathfrak{A}_{R,\infty}$ . In the local normalization used here, its relation to the original block is

$$\boxed{\mathcal{B}_{\text{cur},\beta}^{\eta_3,\eta_2}(z) = \lim_{\substack{c \rightarrow \infty \\ \beta, \beta_2, \beta_3, h_1, h_4 \text{ fixed}}} (1-z)^{c/12} \mathcal{B}_\beta(c; z) = \mathcal{B}_\infty^{\text{red}}(z).}\tag{57}$$

The regular function  $S_\beta(q)$  in the elliptic recursion is this same current-algebra block after the elliptic prefactor and oscillator normalization have been converted from  $z$  to  $q$ ; it is not an independent character-like input.

## 5 All-level evaluation of the local large- $c$ block

Set

$$A = h_1 + h_4 + \beta_2^2 + \beta_3^2.\tag{58}$$

The coherent oscillator sources are awkward in a PBW basis because a nominally subleading mode-changing bracket can multiply an  $O(\sqrt{c})$  source. The way to avoid this order-of-limits problem is to decompose the module into orthogonal global conformal families *before* taking  $c \rightarrow \infty$ .

## 5.1 Exact global-family decomposition

Let

$$\mathcal{Q}_m^{\text{even}} = \left\{ \chi \in \mathcal{W}_{c,\beta,m}^{\text{even}} : L_1 \chi = 0 \right\} \quad (59)$$

be the even Ramond quasi-primary space at level  $m$ . At generic  $(c, \beta)$ , choose a basis  $\{\chi_{m,a}\}_{a=1}^{d_m}$  and write

$$Q_m[a, b] = \langle \chi_{m,a} | \chi_{m,b} \rangle. \quad (60)$$

The standard  $\mathfrak{sl}_2$  decomposition is orthogonal:

$$\mathcal{W}_{c,\beta,N}^{\text{even}} = \bigoplus_{m=0}^N L_{-1}^{N-m} \mathcal{Q}_m^{\text{even}}. \quad (61)$$

Indeed, if  $m > m'$ , moving  $L_1^{N-m}$  through the pairing reduces the cross term to  $\langle \chi_{m,a} | L_{-1}^{m-m'} \chi_{m',b} \rangle = 0$ .

Define the quasi-primary endpoint couplings

$$\begin{aligned} \lambda_{L,m,a} &= \rho_{NR}^{(\eta_3)} \left( \nu_4, w_{\beta_3}^+, \chi_{m,a} \right), \\ \lambda_{R,m,a} &= \rho_{RN}^{(\eta_2)} \left( \chi_{m,a}, w_{\beta_2}^+, \nu_1 \right), \\ K_m(c) &= \sum_{a,b=1}^{d_m} \lambda_{L,m,a} (Q_m^{-1})^{ab} \lambda_{R,m,b}. \end{aligned} \quad (62)$$

The global Ward identities and

$$\langle L_{-1}^k \chi_{m,a} | L_{-1}^k \chi_{m,b} \rangle = k! (2h_R(\beta) + 2m)_k Q_m[a, b]$$

give the exact, finite- $c$  decomposition

$$\boxed{\mathcal{B}_\beta^{\eta_3, \eta_2}(c; z) = \sum_{m=0}^{\infty} z^m K_m(c) \mathcal{G}_{h_R(\beta)+m}(z),} \quad (63)$$

where

$$\mathcal{G}_H(z) = {}_2F_1(H + h_R(\beta_2) - h_1, H + h_R(\beta_3) - h_4; 2H; z). \quad (64)$$

The level-zero quasi-primary is  $\chi_{0,1} = w_\beta^+$ . The normalization of the three-forms gives

$$K_0(c) = 1. \quad (65)$$

## 5.2 Suppression of every positive-level quasi-primary

**Lemma.** For fixed  $\beta, \beta_2, \beta_3, h_1, h_4$ , with  $\beta \neq 0$ , and for every fixed  $m \geq 1$ ,

$$\boxed{K_m(c) = O(c^{-1}).} \quad (66)$$

**Proof.** Normalize the nonzero modes as in Eq. (30), and choose a  $c$ -dependent basis  $\hat{\chi}_{m,a}$  of  $\mathcal{Q}_m^{\text{even}}$  whose Gram matrix has a finite nondegenerate limit. To see that such a basis exists, note that  $L_1 = \sqrt{c/12} a_1$ . Thus the limiting quasi-primary space is the kernel of the Fock annihilator  $a_1$ , namely the span of the contracted basis vectors with no  $a_1^\dagger$  factor. The restriction of Eq. (53) to this span is nondegenerate for  $\beta \neq 0$ . Gaussian elimination in powers of  $c^{-1/2}$  then lifts this limiting basis to  $\ker L_1$  at finite large  $c$ , with a finite nondegenerate Gram limit.

Rotate either N–R–R pair of pants to the R–R presentation using the Möbius map  $f$  of Eq. (43). The internal puncture is fixed by  $f$ . Consequently its local state transport is generated by  $L_0, L_1$  (and the choice of spin lift), with no  $L_{-1}$ . Since  $L_1 \hat{\chi}_{m,a} = 0$ , the transported quasi-primary is therefore just a nonzero scalar multiple of  $\hat{\chi}_{m,a}$ . In particular, none of the lower-level terms responsible for Eq. (56) is produced. This is the exact step that fails if one first works with a PBW descendant instead of projecting to a quasi-primary.

In the resulting R–R presentation, expand  $\hat{\chi}_{m,a}$  in normalized creation modes and move one creation mode at a time through the bottom NS intertwiner. The creation-mode counterparts of Eq. (38) show that each commutator is  $O(c^{-1/2})$  on fixed-level states; the outgoing creation mode itself annihilates the external Ramond ground-state bra. The derivatives in the commutator are finite here: the common  $c/24$  part of the two Ramond weights cancels, while the NS weight and all momenta are fixed. At strict leading order the intertwiner is thus the identity on the nonzero oscillator Fock space, tensored with a finite map on the  $G_0$  ground fibre. A positive-level state contains at least one nonzero creation operator, so induction on their number gives

$$\rho_{L,R}(\hat{\chi}_{m,a}) = O(c^{-1/2}), \quad m \geq 1. \quad (67)$$

There is no hidden coherent enhancement in Eq. (67). Equivalently, within the ground global family one has

$$\langle \hat{\chi}_{m,a} | L_{-1}^m w_\beta^+ \rangle = \langle L_1^m \hat{\chi}_{m,a} | w_\beta^+ \rangle = 0.$$

Thus the  $O(\sqrt{c})$  coherent piece is removed exactly by the global-family projection before the  $c \rightarrow \infty$  limit is taken.

The ordered Ward system gives the same conclusion. If  $|C_{R,N}\rangle$  is the right chain vector, the Virasoro identity is

$$L_n |C_{R,N}\rangle = (h_R(\beta) + N - n + n h_R(\beta_2) - h_1) |C_{R,N-n}\rangle. \quad (68)$$

On the other hand,

$$L_n \frac{L_{-1}^N}{N!} w_\beta^+ = ((n+1)h_R(\beta) + N - n) \frac{L_{-1}^{N-n}}{(N-n)!} w_\beta^+. \quad (69)$$

The difference between the coefficients in Eqs. (68) and (69) is

$$n(h_R(\beta_2) - h_R(\beta)) - h_1 = n(\beta^2 - \beta_2^2) - h_1,$$

which is finite. The two generalized Ramond supercurrent Ward identities give the analogous statement in the fermionic directions: after  $G_n = n\sqrt{c/3}\psi_n$  is extracted, their sources contain only the finite operators  $G_0$  and the fixed external momenta. This is the Ward-identity form of the R–R commutator induction above. Applying the endpoint estimate on both pairs of pants and sewing with the finite inverse quasi-primary Gram matrix proves Eq. (66).  $\square$

The use of the complete even quasi-primary space is important here. The  $G_0$ -ground direction and the fermionic descendants participate in the projection which removes the coherent piece; projecting the ground fibre before forming  $K_m$  would invalidate the lemma.

### 5.3 The surviving ground family

Euler's hypergeometric transformation applied to Eq. (64), with  $H = h_R(\beta) + m$ , gives

$$\begin{aligned} \mathcal{G}_{h_R(\beta)+m}(z) &= (1-z)^{h_1+h_4-h_R(\beta_2)-h_R(\beta_3)} \\ &\quad \times {}_2F_1\left(m+h_1+\beta_2^2-\beta^2, m+h_4+\beta_3^2-\beta^2; \frac{c}{12}-2\beta^2+2m; z\right) \\ &= (1-z)^{-c/12+A} \left[1 + O(c^{-1})\right], \end{aligned} \quad (70)$$

coefficientwise in  $z$ . Notice that the exponent is independent of both the internal momentum and the quasi-primary level  $m$ .

At a fixed power of  $z$ , only finitely many terms of Eq. (63) occur. Combining Eqs. (65), (66), and (70) therefore proves

$$\boxed{\mathcal{B}_\beta^{\eta_3, \eta_2}(c; z) = (1-z)^{-c/12+A} \left[1 + O(c^{-1})\right]}, \quad (71)$$

and hence

$$\boxed{\mathcal{B}_\infty^{\text{red}}(z) = \lim_{c \rightarrow \infty} (1-z)^{c/12} \mathcal{B}_\beta(c; z) = (1-z)^A}. \quad (72)$$

All limits are coefficientwise. The error statement in Eq. (71) is not an absolute estimate on the unstripped coefficients, which grow polynomially in  $c$ .

Equation (72), now a consequence rather than an assumption, is equivalently

$$\left[(1-z)\frac{d}{dz} + A\right] \mathcal{B}_\infty^{\text{red}}(z) = 0, \quad \mathcal{B}_\infty^{\text{red}}(0) = 1. \quad (73)$$

Writing  $\mathcal{B}_\infty^{\text{red}}(z) = \sum_{N \geq 0} C_N z^N$ , the derived recurrence is

$$NC_N + (A - N + 1)C_{N-1} = 0, \quad C_N = (-1)^N \binom{A}{N}. \quad (74)$$

### 5.4 Level-one certificate

In the even basis

$$(L_{-1}w^+, G_{-1}G_0w^+), \quad (75)$$

the Gram matrix is

$$B_{R,1} = \begin{pmatrix} 2h_R(\beta) & -\frac{3}{2}\beta^2 \\ -\frac{3}{2}\beta^2 & -\beta^2(2h_R(\beta) + \frac{c}{4}) \end{pmatrix}. \quad (76)$$

The two Ward vectors are

$$\begin{aligned} \rho_L &= \left(h_R(\beta) + h_R(\beta_3) - h_4, -\frac{\beta^2}{2} - \eta_3\beta\beta_3\right), \\ \rho_R &= \left(h_R(\beta) + h_R(\beta_2) - h_1, -\frac{\beta^2}{2} - \eta_2\beta\beta_2\right). \end{aligned} \quad (77)$$

Their full contraction gives

$$B_1(c) = \rho_L B_{R,1}^{-1} \rho_R^T = \frac{c}{12} - A + O(c^{-1}), \quad (78)$$

which is exactly the first coefficient of  $(1-z)^{-c/12+A}$ . The component signs drop out only after the complete ground-fibre contraction.

## 5.5 The short $\beta = 0$ quotient

The proof above was formulated for a generic long module so that the two-dimensional ground metric is invertible. At  $\beta = 0$  one must not substitute directly into Eq. (54). Instead take the irreducible short quotient

$$G_0 w_0^+ = 0, \quad h_R(0) = \frac{c}{24}, \quad (79)$$

and form its quotient Gram matrices.

The global-family proof simplifies in this quotient. The even ground quasi-primary space is one-dimensional, generated by  $w_0^+$ , and all basis vectors with a rightmost  $G_0 w_0^+$  are absent. The positive-level quotient still contracts to the ordinary boson–fermion Fock metric and has no positive-level vector annihilated by every  $a_n, \psi_n$ . Hence the suppression lemma remains valid:

$$K_m^{\text{short}}(c) = O(c^{-1}), \quad m \geq 1. \quad (80)$$

The surviving ground global family is

$$\begin{aligned} \mathcal{G}_{c/24}^{\text{short}}(z) &= {}_2F_1\left(\frac{c}{12} - \beta_2^2 - h_1, \frac{c}{12} - \beta_3^2 - h_4; \frac{c}{12}; z\right) \\ &= (1-z)^{-c/12+A} {}_2F_1\left(h_1 + \beta_2^2, h_4 + \beta_3^2; \frac{c}{12}; z\right). \end{aligned} \quad (81)$$

Therefore

$$\boxed{\lim_{c \rightarrow \infty} (1-z)^{c/12} \mathcal{B}_0^{\text{short}}(c; z) = (1-z)^A.} \quad (82)$$

This is an independent quotient argument. It does not assert equality between the finite- $c$  long block analytically continued to  $\beta = 0$  and the finite- $c$  short block; subleading terms must be computed with the quotient Gram matrices. The two arguments establish only that their normalized leading large- $c$  term is the same.

## 6 Conversion to the elliptic block

Let

$$q = \exp\left[-\pi \frac{K(1-z)}{K(z)}\right], \quad \Delta_0 = \frac{c - \frac{3}{2}}{24}. \quad (83)$$

Define the elliptic block by the change of normalization

$$\begin{aligned} \mathcal{F}_\beta(z) &= (16q)^{h_R(\beta) - \Delta_0 - \frac{1}{16}} z^{\Delta_0 - h_1 - h_R(\beta_2) + \frac{1}{16}} (1-z)^{\Delta_0 - h_R(\beta_2) - h_R(\beta_3)} \\ &\times \theta_3(q)^{\frac{c - \frac{3}{2}}{2} - 4(h_1 + h_R(\beta_2) + h_R(\beta_3) + h_4) + \frac{1}{2}} \mathcal{H}_\beta(c; q). \end{aligned} \quad (84)$$

At fixed  $\beta$ , the raw  $\mathcal{H}_\beta$  is not bounded as  $c \rightarrow \infty$ . Define

$$P(q) = \prod_{n=1}^{\infty} \frac{1+q^n}{1-q^n}, \quad \mathcal{E}_R(c, q) = P(q)^{c/6}, \quad \hat{\mathcal{H}}_\beta(c; q) = \mathcal{E}_R(c, q)^{-1} \mathcal{H}_\beta(c; q). \quad (85)$$

The product identity

$$\theta_3(q) P(q) = (1 - z(q))^{-1/4} \quad (86)$$

shows that the  $O(c)$  part of Eqs. (84) and (85) is exactly the local factor  $(1-z)^{-c/12}$ .

Combining the proved local limit (72) with Eqs. (84)–(86) gives the normalized all-order regular term:

$$S_\beta(q) = \left( \frac{16q}{z(q)} \right)^{\beta^2} (1 - z(q))^{h_1 + h_4 + \frac{1}{16}} \theta_3(q)^{4(h_1 + h_4 - \beta_2^2 - \beta_3^2) + \frac{1}{4}}. \quad (87)$$

The limiting identity is therefore

$$S_\beta(q) = \lim_{\substack{c \rightarrow \infty \\ \beta, \beta_2, \beta_3, h_1, h_4 \text{ fixed}}} \hat{\mathcal{H}}_\beta(c; q). \quad (88)$$

The result is independent of  $(\eta_3, \eta_2)$ . Its dependence on the internal  $\beta$  comes entirely from the local-coordinate factor  $(16q/z)^{\beta^2}$ .

### 6.1 First two coefficients

Define

$$X = \beta^2 - \beta_2^2 - \beta_3^2 - h_1 - h_4. \quad (89)$$

Expansion of Eq. (87) gives

$$\begin{aligned} S_\beta(q) &= 1 + S_1 q + S_2 q^2 + O(q^3), \\ S_1 &= 8X - \frac{1}{2}, \\ S_2 &= 32X^2 - 16\beta^2 + 12\beta_2^2 + 12\beta_3^2 - 4h_1 - 4h_4 - \frac{3}{8}. \end{aligned} \quad (90)$$

These agree with the independent level-one and level-two Ward–Gram calculations.

## 7 Complete fixed- $\beta$ $c$ -recursion

At each fixed power of  $q$ , subtracting the coefficientwise large- $c$  regular part leaves a rational function of  $c$  that vanishes at infinity. Its partial-fraction expansion is therefore fixed by Eq. (26). As an identity of formal  $q$ -series, the block recursion is

$$\begin{aligned} \hat{\mathcal{H}}_\beta^{\eta_3, \eta_2}(c; q) &= S_\beta(q) \\ &+ \sum_{\substack{r > s \geq 1 \\ r+s \text{ odd}}} \sum_{\alpha = \pm} \frac{q^{rs/2} \mathcal{K}_{r,s}^{(\alpha; \eta_3, \eta_2)}(\beta)}{c - c_{r,s}^{(\alpha)}(\beta)} \hat{\mathcal{H}}_{\beta_{r,s}^{(\alpha)}(\beta)}^{\eta_3, \eta_2} \left( c_{r,s}^{(\alpha)}(\beta); q \right). \end{aligned} \quad (91)$$

Here the external data  $(h_1, h_4, \beta_2, \beta_3)$  are the same in every block on the right-hand side; only the internal momentum is shifted, and the shifted block is evaluated at the pole value of  $c$ . Equation (91) is exact, with the regular term  $S_\beta(q)$  given by Eq. (87).

Write

$$\hat{\mathcal{H}}_\beta(c; q) = \sum_{n=0}^{\infty} \hat{H}_n(c; \beta) q^n, \quad S_\beta(q) = \sum_{n=0}^{\infty} S_n(\beta) q^n. \quad (92)$$

At order  $q^n$ , only labels with  $rs/2 \leq n$  occur. On the canonical positive- $\beta$  sheet the exact coefficient recursion is

$$\hat{H}_n(c; \beta) = S_n(\beta)$$



$$+ \sum_{\substack{r>s\geq 1, \, r+s \text{ odd} \\ rs/2 \leq n}} \sum_{\alpha=\pm} \frac{\mathcal{K}_{r,s}^{(\alpha;\eta_3,\eta_2)}(\beta)}{c - c_{r,s}^{(\alpha)}(\beta)} \hat{H}_{n-rs/2} \left( c_{r,s}^{(\alpha)}(\beta); \beta_{r,s}'^{(\alpha)}(\beta) \right), \quad (93)$$

where the pole position, shifted momentum, and complete residue multiplier are given in Eqs. (19), (20), and (25). The endpoint component labels are unchanged on this canonical sheet.

**Completeness of the pole part.** For generic fixed  $\beta \neq 0$ , the pole sum in Eqs. (91) and (93) is complete, coefficient by coefficient in  $q$ . Indeed:

1. the Ramond Kac determinant exhausts the zeros of the internal Gram determinant, namely  $\beta = \pm \beta_{r,s}(c)$ , with  $r + s$  odd and  $rs/2 \leq n$ ;
2. restricting to  $r > s$  and keeping both  $b_{r,s}^{(\alpha)}$ ,  $\alpha = \pm$ , represents every resulting  $c$ -pole once;
3. the fixed nonzero  $\beta$  ground fibre has  $G_0^2 = -\beta^2$ , so it creates no additional moving level-zero pole;
4.  $C_L C_R$  has been factored out, and the oscillator factor  $\mathcal{E}_R(c, q)$  is entire and nonzero in  $c$ , so neither alters the conformal-block pole set.

Fusion-polynomial zeros can cancel individual residues but cannot produce new poles. At coincident or ramified Kac loci the displayed simple-pole sum is understood by analytic continuation from generic data.

Together with the all-order regular term Eq. (87), this completeness statement closes the fixed- $\beta$   $c$ -recursion coefficient by coefficient.

## 8 Checks and present status

### 8.1 Analytic direct truncation through level four

The direct check uses the complete even PBW spaces, whose dimensions at levels zero through four are

$$1, \quad 2, \quad 4, \quad 8, \quad 14.$$

For reference, ordered bases may be chosen as

$$\begin{aligned} \mathcal{H}_0^{\text{even}} &: \{w\}, \\ \mathcal{H}_1^{\text{even}} &: \{L_{-1}w, G_{-1}G_0w\}, \\ \mathcal{H}_2^{\text{even}} &: \{L_{-1}^2w, L_{-2}w, G_{-2}G_0w, L_{-1}G_{-1}G_0w\}, \\ \mathcal{H}_3^{\text{even}} &: \{L_{-1}^3w, L_{-2}L_{-1}w, L_{-3}w, G_{-3}G_0w, L_{-1}G_{-2}G_0w, L_{-1}^2G_{-1}G_0w, \\ &\quad L_{-2}G_{-1}G_0w, G_{-2}G_{-1}w\}, \\ \mathcal{H}_4^{\text{even}} &: \{L_{-1}^4w, L_{-2}L_{-1}^2w, L_{-2}^2w, L_{-3}L_{-1}w, L_{-4}w, G_{-4}G_0w, \\ &\quad L_{-1}G_{-3}G_0w, L_{-1}^2G_{-2}G_0w, L_{-1}^3G_{-1}G_0w, L_{-2}G_{-2}G_0w, \\ &\quad L_{-2}L_{-1}G_{-1}G_0w, L_{-3}G_{-1}G_0w, G_{-3}G_{-1}w, L_{-1}G_{-2}G_{-1}w\}. \end{aligned}$$

Let  $B_{R,N}$  be the Gram matrix and let  $\boldsymbol{\rho}_{L,N}, \boldsymbol{\rho}_{R,N}$  be the two Ward vectors in these bases. The exact level-four truncation is the analytic expression

$$\mathcal{B}_\beta^{[4]}(c; z) = 1 + \sum_{N=1}^4 z^N \boldsymbol{\rho}_{L,N} B_{R,N}^{-1} \boldsymbol{\rho}_{R,N}^\top + O(z^5). \quad (94)$$

This involves only finite matrices of sizes 2, 4, 8, 14; no pole recursion enters.

It is important that Eq. (94) is the finite- $c$  superconformal block itself, before any large- $c$  normalization. If

$$F_N(c) = \boldsymbol{\rho}_{L,N} B_{R,N}^{-1} \boldsymbol{\rho}_{R,N}^\top,$$

then an entirely scalar analytic form of the same coefficient is the bordered-determinant formula

$$F_N(c) = - \frac{\det \begin{pmatrix} 0 & \boldsymbol{\rho}_{L,N} \\ \boldsymbol{\rho}_{R,N}^\top & B_{R,N} \end{pmatrix}}{\det B_{R,N}}. \quad (95)$$

Thus the full four-point block, including its OPE power, is

$$\boxed{\mathcal{F}_\beta^{\eta_3, \eta_2, [4]}(c; z) = z^{-\beta^2 + \beta_2^2 - h_1} \left( 1 + F_1(c)z + F_2(c)z^2 + F_3(c)z^3 + F_4(c)z^4 + O(z^5) \right)}. \quad (96)$$

All entries of the four Gram matrices and the eight Ward vectors are polynomials in

$$c, \beta, \beta_2, \beta_3, h_1, h_4, \eta_2, \eta_3.$$

They are recorded in the accompanying exact symbolic data file `ramond_sphere_level4_exact_expansion.md`. This is a machine-readable form of Eq. (96), not a numerical implementation. For example, at level one,

$$\begin{aligned} B_{R,1} &= \begin{pmatrix} \frac{c}{12} - 2\beta^2 & -\frac{3}{2}\beta^2 \\ -\frac{3}{2}\beta^2 & 2\beta^4 - \frac{c}{3}\beta^2 \end{pmatrix}, \\ \boldsymbol{\rho}_{L,1} &= \left( \frac{c}{12} - \beta^2 - \beta_3^2 - h_4, -\frac{1}{2}\beta^2 - \eta_3\beta\beta_3 \right), \\ \boldsymbol{\rho}_{R,1} &= \left( \frac{c}{12} - \beta^2 - \beta_2^2 - h_1, -\frac{1}{2}\beta^2 - \eta_2\beta\beta_2 \right). \end{aligned}$$

Substitution in Eq. (95) gives the complete finite- $c$  coefficient  $F_1(c)$ ; the higher-level entries are obtained in exactly the same way from the displayed PBW bases.

Only after the finite- $c$  block has been constructed in this way do we compare it with the contracted block. Define the coefficients

$$C_N(c) = \sum_{k=0}^N (-1)^k \binom{c/12}{k} \boldsymbol{\rho}_{L,N-k} B_{R,N-k}^{-1} \boldsymbol{\rho}_{R,N-k}^\top. \quad (97)$$

Here the level-zero contraction is one. Directly evaluating the finite Ward vectors and Gram matrices and then taking  $c \rightarrow \infty$ , with all data in Eq. (2) held fixed, gives

$$\begin{aligned} \lim_{c \rightarrow \infty} C_0(c) &= 1, \\ \lim_{c \rightarrow \infty} C_1(c) &= -A, \\ \lim_{c \rightarrow \infty} C_2(c) &= \frac{A(A-1)}{2}, \\ \lim_{c \rightarrow \infty} C_3(c) &= -\frac{A(A-1)(A-2)}{6}, \\ \lim_{c \rightarrow \infty} C_4(c) &= \frac{A(A-1)(A-2)(A-3)}{24}. \end{aligned} \quad (98)$$

All dependence on  $\beta, \eta_2, \eta_3$  cancels in these five direct contractions. Thus the analytic level-four descendant calculation is

$$\boxed{\lim_{c \rightarrow \infty} (1-z)^{c/12} \mathcal{B}_\beta^{[4]}(c; z) = 1 - Az + \frac{A(A-1)}{2} z^2 - \frac{A(A-1)(A-2)}{6} z^3 + \frac{A(A-1)(A-2)(A-3)}{24} z^4 + O(z^5).} \quad (99)$$

This is precisely the truncation of  $(1-z)^A$ , but it has been obtained here from the finite super-Virasoro Ward–Gram sewing.

## 8.2 Other checks

The fixed- $\beta$  recursion and the independent direct descendant implementation perform the following checks:

1. the two level-one  $c$ -poles agree with the zeros of the exact  $2 \times 2$  Ramond Gram determinant;
2. the residue multipliers agree with direct Laurent extraction from the Ward–Gram block at the simple Kac poles;
3. Eq. (87) reproduces the independently derived  $S_1$  and  $S_2$ ;
4. direct PBW/Gram/Ward sewing along a fixed- $\beta$   $c$ -family agrees with the closed  $c$ -recursion through level five, for all four choices of  $(\eta_3, \eta_2)$ ;
5. after multiplication by  $(1-z)^{c/12}$ , the same direct sewing converges through level five to the coefficients of  $(1-z)^A$ , with the component dependence vanishing as  $O(c^{-1})$ .

The checks are independent low-order certificates for the global-family derivation. In particular, they verify explicitly that the  $G_0$ -ground-fibre and sign-dependent contributions combine into the  $O(c^{-1})$  positive-quasi-primary remainder predicted by Eq. (66).